

DA6360 - WS. 4

1. $x_* = 0$

$$\lim_{t \rightarrow \infty} \frac{|x_{t+1}|}{|x_t|^q} = \mu$$

a) $x_t = \frac{1}{t}$ $\frac{x_{t+1}}{x_t^q} = \frac{\frac{1}{t+1}}{\left(\frac{1}{t}\right)^q} = \frac{t^q}{t+1}$
 $t \rightarrow \infty \quad \frac{t^q}{t+1} \sim \frac{t^{q-1}}{1}$

$q < 0 \Rightarrow \text{limit} = 0$

$q = 1 \Rightarrow \text{limit} = 1$

$q > 1 \Rightarrow \text{limit} = +\infty$

$q = 1$ (order), $\mu = 1$ (rate)

b) $x_t = \frac{1}{t^2}$ $\frac{x_{t+1}}{x_t^q} = \frac{\frac{1}{(t+1)^2}}{\left(\frac{1}{t^2}\right)^q} = \frac{t^{2q}}{(t+1)^2}$
 $t \rightarrow \infty \quad \frac{t^{2q}}{(t+1)^2} \sim t^{2q-2}$

$q < 0 \Rightarrow \text{limit} = 0$

$q = 1 \Rightarrow \text{limit} = 1$

$q > 1 \Rightarrow \text{limit} = +\infty$

$q = 1$ (order), note $\Rightarrow \mu = 1$

$$(c) x_t = \frac{1}{2^t} \quad \frac{x_{t+1}}{x_t^q} = \frac{2^{-(t+1)}}{(2^{-t})^q} = 2^{-t-1+qt} = 2^{(q-1)t-1}$$

$$q < 1 \Rightarrow \text{limit} = 0$$

$$q = 1 \Rightarrow \text{limit} = 2^{-1} = \frac{1}{2}$$

$$q > 1 \Rightarrow \text{limit} = +\infty$$

$$q = 1, \quad \mu = \frac{1}{2}$$

$$(d) x_t = \frac{1}{2^{t^2}} \quad \frac{x_{t+1}}{x_t^q} = \frac{2^{-(t+1)^2}}{(2^{-t^2})^q} = 2^{-(t+1)^2 + qt^2} = 2^{(q-1)t^2 - 2t - 1}$$

$$q < 1 \Rightarrow \text{limit} = 0$$

$$q = 1 \Rightarrow \text{limit} = 0$$

$$q > 1 \Rightarrow \text{limit} = +\infty$$

no constant q for which the limit is a finite non-0 no.

$$(e) x_t = \frac{1}{2^{2^t}} \quad \frac{x_{t+1}}{x_t^q} = \frac{2^{-2^{t+1}}}{(2^{-2^t})^q} = 2^{-2^{t+1} + q \cdot 2^t} = 2^{-2^t(2 - q)}$$

$$= 2^{(q-2)2^t}$$

$$q < 2 \Rightarrow \text{limit} = 0$$

$$q = 2 \Rightarrow \text{limit} = 1$$

$$q > 2 \Rightarrow \text{limit} = +\infty$$

$$q = 2, \mu = 1$$

$$2. \quad e_t = x_t - x_*$$

$$|e_{t+1}| = |e_t|^q \quad (q \geq 2)$$

$$e_0 = 0.1 \quad \& \quad \varepsilon > 0$$

find min. t s.t. $e_t \leq \varepsilon$

$$a_{t+1} = a_t^q, \quad q \geq 2, \quad a_0 = |e_0| = 0.1$$

$$a_1 = (0.1)^q, \quad a_2 = (0.1)^{q^2}$$

$$a_t = (0.1)^{q^t}$$

$$(0.1)^{q^t} \leq \varepsilon$$

$$= 10^{-q^t} \leq \varepsilon$$

$$-q^t \leq \log_{10} \varepsilon$$

$$q^t \geq \log_{10} \left(\frac{1}{\varepsilon} \right)$$

$$t \geq \log_2 \left(\log_{10} \frac{1}{\epsilon} \right)$$

$$t_{\min} = \left\lceil \log_2 \left(\log_{10} \frac{1}{\epsilon} \right) \right\rceil$$

3.

$$f(x) = ax^2 + bx + c$$

$$a \neq 0$$

$$x_{t+1} = x_t - \alpha f'(x_t)$$

$$f'(x) = 2ax + b$$

$$x_{t+1} = x_t - \alpha (2ax_t + b)$$

minimises

$$f'(x) = 2ax_* + b = 0$$

$$x_* = -\frac{b}{2a}$$

error

$$l_t = x_t - x_*$$

$$x_{t+1} = x_t - \alpha (2ax_t + b)$$

$$2ax_t + b = 2a(x_t - x_*)$$

$$x_{t+1} = x_t - 2a\alpha (x_t - x_*)$$

$$l_{t+1} = (1 - 2a\alpha) l_t$$

closed form

$$e_t = (1 - 2a\alpha)^t e_0$$

$$x_t - x_* = (1 - 2a\alpha)^t (x_0 - x_*)$$

$$x_t + \frac{b}{2a} = (1 - 2a\alpha)^t \left(x_0 + \frac{b}{2a} \right)$$

$$x_t = x_* + (1 - 2a\alpha)^t (x_0 - x_*)$$

allowed range

$$0 < \alpha < \frac{1}{a}$$

$$4. \quad x_{k+1} = x_k - \alpha f'(x_k)$$

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$$

$$(a) \quad f(x) = \begin{cases} x^2 & |x| \leq 0.1 \\ x^4 + (0.1)^2 - (0.1)^4 & \text{o.w.} \end{cases}$$

$$x^* = 0$$

$$\begin{array}{l} |x| \leq 0.1 \\ \text{o.w.} \end{array} \quad \begin{array}{l} f'(x) = 2x, \quad f''(x) = 2 \\ f'(x) = 4x^3, \quad f''(x) = 12x^2 \end{array}$$

$$x_{k+1} = \begin{cases} x_k - 2\alpha x_k & |x_k| \leq 0.1 \\ x_k - 4\alpha x_k^3 & |x_k| > 0.1 \end{cases}$$

local convergence to 0 for $0 < \alpha < 1$

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$$

$$|x_k| \leq 0.1$$

$$x_{k+1} = x_k - \frac{2x_k}{2} = 0$$

$$|x_k| > 0.1$$

$$x_{k+1} = x_k - \frac{4x_k^3}{12x_k^2} = \frac{2}{3}x_k$$

$$x_{k+1} = \begin{cases} 0 & |x_k| \leq 0.1 \\ \frac{2x_k}{3} & |x_k| > 0.1 \end{cases}$$

$$(b) \quad \begin{aligned} f(x) &= |x|^3 \\ f'(x) &= 3x|x| & f''(x) &= 6|x| \end{aligned}$$

$$x_{k+1} = x_k - \frac{3x_k|x_k|}{6|x_k|}$$

$$x_{k+1} = x_k - \frac{\cancel{3} x_k \cancel{|x_k|}}{\cancel{2} \cancel{6} \cancel{|x_k|}} = \frac{x_k}{2}$$

$$(c) \quad f(x) = \frac{1}{3}|x|^3 - |x|$$

$$x > 0$$

$$f(x) = \frac{1}{3} x^3 - x \quad f'(x) = x^2 - 1$$

$$f''(x) = 2x$$

$$x < 0$$

$$f(x) = -\frac{1}{3} x^3 + x \quad f'(x) = 1 - x^2$$

$$f''(x) = -2x$$

$$x_k > 0$$

$$x_{k+1} = x_k - \alpha (x^2 - 1)$$

$$x_k < 0$$

$$x_{k+1} = x_k - \alpha (1 - x^2)$$

$$\begin{aligned} x_{k+1} &= x_k - \frac{f'(x_k)}{f''(x_k)} \\ &= \frac{x_k^2 + 1}{2x_k} \end{aligned}$$

$$(d) \quad f(x) = x^{4/5}$$

$$f'(x) = \frac{4}{5} \operatorname{sign}(x) |x|^{-1/5}$$

$$f''(x) = -\frac{4}{25} |x|^{-6/5}$$

$$x_{k+1} = x_k - \alpha \frac{4}{5} \operatorname{sign}(x) |x_k|^{-1/5}$$

$$x_{k+1} = x_k + 5x_k = 6x_k$$